Parity Blindness of Finite-Window Weil Detectors at a Real Zero: the Even Sector Cannot See a Siegel Configuration and the Odd Sector Is Maximally Sensitive to It

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Abstract

In the finite-window Weil form, a real zero pair $\{\frac{1}{2}+\delta, \frac{1}{2}-\delta\}$ contributes $2F(i\delta)F(-i\delta)$, and because the autocorrelation factorises this is $+2A_\delta(f)^2$ for even test functions and $-2S_\delta(f)^2$ for odd ones, with $A_\delta = \int f\cosh(\delta u)\,du$ and $S_\delta = \int f\sinh(\delta u)\,du$. The perturbation is therefore rank one, positive semidefinite on the even sector and negative semidefinite on the odd sector, for every window length and in every basis. Two consequences follow. Even-sector finite-window positivity is compatible with a real zero at every $\delta \in (0, \frac{1}{2})$, so no even-sector positivity computation — at any L, in any basis, to any precision — carries information about the Siegel configuration; the window is not merely blind there, it is anti-detecting, and the statement is exact in δ rather than a vanishing leading coefficient. Conversely the odd-sector sensitivity at $\gamma = 0$ is the γ-uniform maximum $2 \cdot (2L^3/3)$ of the identity $K_{\text{even}}(L,\gamma) + K_{\text{odd}}(L,\gamma) = 2L^3/3$, so $\gamma = 0$ is the unique ordinate at which the two sectors' roles separate completely. The sign structure is verified in 150 cells to 8.49×10^{-61}, reproduces the repository's independently derived odd-sector pole term entrywise with no normalisation adjustment, and holds exactly over $\mathbb{Q}$ in the function-field control. This resolves non-claim 9 of the companion note [3]. **The phenomenon is not new:** it is contained in Bombieri [1, §9, Thm. 9], as a count of negative eigenvalues by parity in which a real zero appears as a conjugate pair with $\Re(\gamma) = 0$ and is counted in the odd sector only. No priority is claimed for it. What is offered here is the explicit rank-one form, exact in δ, together with the extremal identity, the two checks, and the consequence for finite-window detectors.

0. Non-claims

Stated first.

1. **Nothing about RH or GRH**, and **nothing about whether Siegel zeros exist.** The theorem below describes what a window form does when a real zero is placed in the zero multiset by hand. It is a statement about a detector, not about the zeros of any L-function.
2. **The quasi-Riemann-hypothesis and Siegel-exclusion preprints of October 2026 are not cited as established and are not used anywhere.** No result here depends on them, and none is asserted about them.

3. **Every planted detection is [T2-N].** A planted real zero is a perturbation of a computed matrix, not an arithmetic fact, and **non-detections certify nothing**.

4. **"The odd sector detects by $L = 1.4$" is [T2-N] twice over.** It is a numerical statement about a planted configuration, and the odd-sector *baseline* it is measured against has no certified lower bound for any Dirichlet window. (For ζ one now exists, including in the odd sector [6, Thm. 6.2]; see §5.) A certificate of that kind would in any case **not** convert the measurement into an exclusion of a real zero — §5 explains why not, and names the ingredient that would. This note supplies neither.

5. **The parent extension's Dirichlet-character results beyond its §5 are not used here.** Saturation ratios, prime-side coincidences, conductor-versus-height observations and the Phase-4 mechanism hypothesis play no part in anything below.

6. **Corollary 2 bounds what a method can detect, not what exists.** Section 5 is a scope statement about even-sector techniques. It is not a criticism of any result, and no cited work claims otherwise.

7. **The function-field section is a control, not a transfer.** In the function field the analogue of RH is Weil's theorem, so there is no configuration there to exclude.

Epistemic tiers

Tier	Meaning	Discharged by
T1	exact or proved	a proof, or exact arithmetic over $\mathbb{Z}$ or $\mathbb{Q}$
T2	numerical	stated working precision + a convergence or noise-floor check
T2-C	certified arithmetic	an interval/ball enclosure with a reproducible certificate
T2-N	high-precision numerical	stable under precision increase, not an enclosure
T3	interpretation	argument from T1/T2 evidence, labelled as such

Every number printed in this note is re-derived from the extension's data files by `note/verify_note_numbers.py`, which also fails if the note prints more significant digits than its source carries.

1 Setup

Let g be real and supported in $[-L, L]$. Following the companion note [3, §2], put

$$G(z) = \int g(u)e^{zu}\,du, \qquad h(v) = \int g(u+v)g(u)\,du, \qquad H(z) = G(z)G(-z),$$

so h is supported in $[-2L, 2L]$ and $H(it) = |G(it)|^2$ for real g. The spectral side of the explicit formula runs over the zeros of the completed function, and all reported values are normalised as $Q(g)/\|g\|_2^2$. For ζ the window form is this repository's assembled Q_ζ; for a real primitive χ modulo q it is

$$M_\chi = \mathrm{Arch}_a - \mathrm{LogPi} + \mathrm{LogQ} - \mathrm{Prime}_\chi,$$

with no pole term, χ being non-principal. Both parity sectors are used throughout: the even bases $\cos((2k+1)\pi x/2L)$ and $\cos(k\pi x/L)$, and the odd basis $\sin(k\pi x/L)$, all normalised.

For a non-real off-line zero the complete functional-equation and conjugation orbit is $\mathcal{O} = \{\beta \pm i\gamma,\ 1-\beta \pm i\gamma\}$, and its contribution is [3, eq. (4)]

$$Q_{\mathcal{O}}(g) = 4\,\mathrm{Re}\Big(A\overline{B}\Big) = 4\left(|P|^2 - |S|^2\right), \qquad A = G(\delta+i\gamma), \quad B = G(-\delta+i\gamma), \tag{1}$$

with $P = \int g\cosh(\delta u)e^{i\gamma u}du$ and $S = \int g\sinh(\delta u)e^{i\gamma u}du$, $\delta = \beta - \frac{1}{2}$: a signed rank-two form, positive on the cosh channel and negative on the sinh channel. The present note is the $\gamma = 0$ corner of (1), where the orbit degenerates to a *pair* and the two channels separate completely by parity.

2 The theorem

Theorem 1 (sign structure at a real zero)**.** *[T1] Let f be real and supported in $[-L, L]$, and let a real zero pair $\{\frac{1}{2} + \delta,\ \frac{1}{2} - \delta\}$, $\delta \in (0, \frac{1}{2}]$, be present in the zero multiset. (The open interval $\delta \in (0, \frac{1}{2})$ is the case of a* zero*; the endpoint $\delta = \frac{1}{2}$ is the* pole*, treated separately in Remark 6 and verified as such in §3(c).) Its contribution to the finite-window Weil form is*

$$2\left(\int f(u)e^{\delta u}du\right)\left(\int f(u)e^{-\delta u}du\right) = \begin{cases} +2\,A_\delta(f)^2, & f \text{ even}, \quad A_\delta(f) = \displaystyle\int f(u)\cosh(\delta u)\,du, \\ -2\,S_\delta(f)^2, & f \text{ odd}, \quad S_\delta(f) = \displaystyle\int f(u)\sinh(\delta u)\,du, \end{cases}$$

for every $L > 0$ and in every basis. In particular the perturbation is rank one, positive semidefinite on the even sector and negative semidefinite on the odd sector.

Proof. Four lines; written out with every step in Appendix A. The zero side is $\sum_\rho h(z_\rho)$ with $h(z) = F(z)F(-z)$, $\rho = \frac{1}{2} + iz$, and the autocorrelation *factorises* — that is the whole content. The functional equation forces the partner, so the pair sits at $z = \mp i\delta$ and contributes $h(i\delta) + h(-i\delta) = 2h(i\delta) = 2F(i\delta)F(-i\delta)$. With $F(i\delta) = \int f(x)e^{-\delta x}dx$ and $e^{-\delta x} = \cosh(\delta x) - \sinh(\delta x)$, parity gives $F(\pm i\delta) = (A_\delta, A_\delta)$ for even f and $(-S_\delta, +S_\delta)$ for odd f. Even f gives $A \cdot A$, odd f gives $S \cdot (-S)$. The sign is not a convention: it is the parity of F at a purely imaginary argument. □

Corollary 2 (the even sector is anti-detecting)**.** *[T1] Suppose every non-trivial zero of $L(s, \chi)$ lies on the critical line except a single real pair $\frac{1}{2} \pm \delta$. Then for every $L > 0$ and every real even f supported in $[-L, L]$,*

$$Q_\chi(f) = \sum_{\text{on-line } \rho} |F(\gamma_\rho)|^2 + 2A_\delta(f)^2 \ \geq\ 2A_\delta(f)^2 \ \geq\ 0.$$

Hence even-sector finite-window Weil positivity is compatible with a real zero at every $\delta \in (0, \frac{1}{2})$, and no even-sector positivity computation — at any window length, in any basis, to any precision — carries information about the Siegel configuration. Equivalently, as a perturbation of the matrix: the response is $+2aa^{\mathsf{T}}$ with $a_k = \int \varphi_k \cosh(\delta u)\,du/\|\varphi_k\|$, so $\lambda_{\min}(Q + 2aa^{\mathsf{T}}) \geq \lambda_{\min}(Q)$ for every basis and every N.

Remark 3. [T1] This is stronger than "the δ^2 coefficient vanishes". That weaker statement is true — $K(L, 0) = \int_{-L}^{L} u^2 \sin^2(0 \cdot u)du = 0$, i.e. $F'(0) = 0$ for even f — but it only kills the *leading* coefficient. Theorem 1 is exact in δ: the response is a perfect square of the wrong sign to all orders. The even sector is not blind at $\gamma = 0$; a planted real zero makes the form *more* positive.

Corollary 4 (the odd sector is maximally sensitive)**.** *[T1] For real odd f the same identity reads $Q_\chi(f) = \sum_{\text{on-line}} |\widetilde{F}(\gamma_\rho)|^2 - 2S_\delta(f)^2$, which can be negative. Moreover the derivative-evaluation extremal in the odd sector at $\gamma = 0$ is*

$$2\sup\left\{\left(\int_{-L}^{L} u\,g(u)\,du\right)^2 : g \text{ odd},\ \|g\|_2 = 1\right\} = 2\,K_{\text{odd}}(L, 0) = 2 \cdot \frac{2L^3}{3},$$

attained at $g \propto u$, which is the γ-uniform maximum *of the identity $K_{\text{even}}(L, \gamma) + K_{\text{odd}}(L, \gamma) = 2L^3/3$ together with $K_{\text{even}}(L, 0) = 0$.*

So $\gamma = 0$ is a blind spot of one *sector*, not of the window, and the complementary sector is there as sensitive as it ever gets at any ordinate.

Remark 5 (general f: lowering the form needs an odd component). [T1] For real $f = f_{\mathrm{e}} + f_{\mathrm{o}}$ the cross integrals vanish by parity, so the computation of Appendix A gives $F(i\delta) = A_\delta(f_{\mathrm{e}}) - S_\delta(f_{\mathrm{o}})$ and $F(-i\delta) = A_\delta(f_{\mathrm{e}}) + S_\delta(f_{\mathrm{o}})$, and the real pair contributes $2(A_\delta(f_{\mathrm{e}})^2 - S_\delta(f_{\mathrm{o}})^2)$. A real pair can therefore lower the form only through a test function with a nonzero odd component, and only when $|S_\delta(f_{\mathrm{o}})| > |A_\delta(f_{\mathrm{e}})|$. The statements below are in this sense about the odd sector itself and not about a choice of basis.

Remark 6 (the $\delta = \frac{1}{2}$ instance is independently in print). [T1] The $\delta = \frac{1}{2}$ case of Theorem 1, together with Remark 5, is Zhu's parity splitting [6, Lem. 6.1]: for real $f = f_{\mathrm{e}} + f_{\mathrm{o}}$ the pole term is $+2(\int f_{\mathrm{e}} \cosh(x/2))^2$ on the even part and $-2(\int f_{\mathrm{o}} \sinh(x/2))^2$ on the odd part, so "the pole term changes sign in the odd sector", with the same factorisation and the same absence of a cross term. That was derived independently, is in print first, and is validated numerically there to relative error 2×10^{-11}. What Theorem 1 adds is the parameter: the same algebra at $\delta \in (0, \frac{1}{2})$ describes a *real zero* rather than the *pole*, which is what makes it a statement about the Siegel configuration. No priority is claimed for the parity splitting itself, and §3(c) verifies the $\delta = \frac{1}{2}$ identification from the other side.

Prior and concurrent work: the phenomenon is Bombieri's. [T1] The sign-by-parity phenomenon of Theorem 1 is contained in Bombieri [1, §9, Thm. 9, p. 217], in the form of a count rather than a formula. His §9 is titled "the second eigenvalue problem: even and odd eigenfunctions" and splits the problem on a symmetric E by $K_E^{\pm}(x,y) = \int_E e^{i(x-y)u} du \pm \int_E e^{i(x+y)u} du$. Theorem 9 then counts negative eigenvalues: those of K_E^+ are the distinct complex conjugate pairs with $\Re(\gamma) > 0$, those of K_E^- the pairs with $\Re(\gamma) \geq 0$. In his coordinates $\rho = \frac{1}{2} + i\gamma$, a *real* zero $\rho = \frac{1}{2} + \delta$ is $\gamma = -i\delta$, purely imaginary, so $\Re(\gamma) = 0$: **such a pair is counted by the odd sector and by the even sector not at all.** His proof exhibits the mechanism and not only the count — for $\gamma_0 = ib$ purely imaginary, $K^{\pm}(\gamma_0, \gamma_0) = \int_E (1 \pm e^{-2bu}) du \sim \pm B$, so the even sector gains a positive direction and the odd sector a negative one, from the same $1 \pm e^{-2bu}$ split that produces the cosh and sinh above [1, p. 218].

This was not known to the author when version 1 was deposited, and **no priority is claimed for the phenomenon.** What this note adds is the explicit form and its consequences: the rank-one perturbation $+2A_\delta^2$ / $-2S_\delta^2$ with a four-line proof, *exact in* δ rather than a count of negative directions; the extremal identity $K_{\mathrm{even}} + K_{\mathrm{odd}} = 2L^3/3$ with the odd-sector maximum at $\gamma = 0$ (Corollary 4); the sign verified exactly over $\mathbb{Q}$ in the function field (§4); the entrywise agreement with the independently derived pole term at $\delta = \frac{1}{2}$ (§3(c)); the transposition to Dirichlet L-functions; and the consequence stated for finite-window *detectors* (§5). All of that stands; only the implied novelty of the mechanism changes.

$\gamma = 0$ is the only ordinate where the sectors separate. The identity $K_{\mathrm{even}} + K_{\mathrm{odd}} = 2L^3/3$ holds for every γ, but the split is extreme only at $\gamma = 0$. At large γ the two are nearly equal: the companion note measures $K_{\mathrm{even}}/K_{\mathrm{odd}} = 1.00634$ at $L = 2$ and $\gamma = \gamma_1$, the first off-line ordinate of the Davenport–Heilbronn function, $\gamma_1 \approx 85.7$ [3, §6]. The complete separation of roles is a $\gamma = 0$ phenomenon.

Verified at $\gamma = 0$, with K_{odd} obtained by quadrature rather than by subtraction:

L	$K_{\mathrm{even}}(L,0)$	$K_{\mathrm{odd}}(L,0)$	$2L^3/3$	$\lvert\mathrm{err}\rvert$
0.5	0	0.0833333333333	0.0833333333333	0
0.8	0	0.341333333333	0.341333333333	0
1.0	0	0.666666666667	0.666666666667	0
1.3	0	1.46466666667	1.46466666667	0
2.0	0	5.33333333333	5.33333333333	0

3 Verification

Three independent checks, from `src/verify_siegel_parity.py`. All three passed, and were re-run for this note and reproduced the committed transcript byte for byte.

(a) The planted object was the real *pair*. Decided from the committed Phase-5 data rather than from a reading of the code, by recomputing stored eigenvalues both ways. The degenerate quartet is exactly twice the pair — $\max|\text{quartet}(\gamma = 0) - 2 \cdot \text{pair}| = 4.98 \times 10^{-60}$ — so the two hypotheses differ by a factor of 2 and the data distinguishes them: all twelve probed rows are reproduced by the PAIR, unanimously. **No factor-of-2 correction is needed anywhere in this note.** [T2-N]

(b) Rank one, correct sign, eigenvalue $= +2A^2$ / $= -2S^2$. 5 characters $\times$ $L \in \{0.8, 1.0, 1.4\}$ $\times$ 2 sectors $\times$ $\delta \in \{0.1, 0.25, 0.375, 0.45, 0.49\}$, $N = 10$: 150 cells, 150 with the predicted sign. A and S are computed by quadrature on the explicit normalised eigenfunction, so this is an independent check of the closed forms, not a restatement of them. [T2-N]

quantity	worst over the 150 cells
relative error, eigenvalue vs $+2A^2$ (even) / $-2S^2$ (odd)	8.49×10^{-61}
rank-one residual $\max_{ij} \lvert B_{ij} - \lambda v_i v_j \rvert$	2.49×10^{-60}
largest non-dominant eigenvalue	1.45×10^{-60}
the complementary integral (S on even f, A on odd f)	exactly 0
even-sector eigenvalues, 75 of 75	$+3.14163 \ \ldots \ +6.38933$
odd-sector eigenvalues, 75 of 75	$-0.905634 \ \ldots \ -0.00643962$
$K_{\text{even}} + K_{\text{odd}} = 2L^3/3$, K_{odd} by quadrature	1.56×10^{-61}

L	δ	$+2A_\delta^2$ (even)	$-2S_\delta^2$ (odd)	rank-one resid.	rel. err.
0.8	0.1	3.14162843133	−0.00643961708992	6.22×10^{-61}	8.49×10^{-61}
0.8	0.49	3.29375936732	−0.159042352281	3.69×10^{-61}	1.89×10^{-61}
1.0	0.1	3.93155834205	−0.0125860661447	6.42×10^{-61}	1.58×10^{-61}
1.0	0.49	4.23383664019	−0.315813694881	6.22×10^{-61}	2.46×10^{-61}
1.4	0.1	5.52110947823	−0.0345998225163	6.22×10^{-61}	5.64×10^{-61}
1.4	0.49	6.38932656805	−0.905634273488	6.22×10^{-61}	2.58×10^{-61}

(The extreme δ only; all 150 cells are in `data/verify_siegel_parity.csv`.)
The matrices do not depend on χ at all. Grouping the 150 cells by $(L, \text{sector}, \delta)$ leaves exactly *one* distinct eigenvalue in each of the 30 groups: the response to a real zero is a property of the basis and δ only. That is why the conclusion is uniform over the family rather than five separate measurements, and it is why the table above has no χ column.

(c) Pole consistency — the strongest check in this note. The $\delta = \frac{1}{2}$ real-pair perturbation was compared entrywise with this repository's ζ-side pole term $2\,\text{pole_sign}(\text{sector}) P_j P_k / \| \cdot \|$, reporting *both* differences so that the data fixes the sign. [T2-N]

sector	L	pole sign	$\max\lvert B - \text{Pole}\rvert$	$\max\lvert B + \text{Pole}\rvert$	vanishes
EVEN	0.5	+1	1.07×10^{-61}	4.084	B − Pole
EVEN	0.8	+1	3.11×10^{-61}	6.749	B − Pole
EVEN	1.3	+1	3.6×10^{-61}	11.95	B − Pole
EVEN_D	0.5	+1	1.94×10^{-62}	3.281	B − Pole
EVEN_D	0.8	+1	3.11×10^{-61}	5.347	B − Pole
EVEN_D	1.3	+1	6.22×10^{-61}	9.128	B − Pole
ODD	0.5	−1	1.22×10^{-63}	0.05108	B − Pole
ODD	0.8	−1	9.72×10^{-63}	0.2119	B − Pole
ODD	1.3	−1	1.94×10^{-62}	0.9408	B − Pole

B = Pole exactly in all 9 cells, with $\max|B - \text{Pole}| \le 6.22 \times 10^{-61}$, while the alternative $\max|B + \text{Pole}|$ is 0.05108 to 11.95 — wrong by $O(1)$ to $O(10)$, so the test is decisive. **There is no normalisation mismatch: no factor of 2 and no factor of $\frac{1}{2}$.** This is the strongest check because the pole term was derived independently and earlier, for a different purpose, in this repository's ζ assembly, and Theorem 1 reproduces it as the $\delta = \frac{1}{2}$ instance of $-2S_\delta^2$ in the odd sector and of $+2A_\delta^2$ in the even ones.

Remark 7 (a sign that must not be misread). The odd sector's pole_sign is -1, so the odd-sector pole term carries the sign *opposite to the even-sector one*, and the real-pair perturbation reproduces that flip exactly. It is **not** opposite in sign to the pole term itself. The distinction matters because Corollary 2 rests on this sign.

The χ-analogue is identically zero, and that is the point. M_χ reconstructs as Arch $-$ LogPi $+$ LogQ $-$ Prime with residual exactly 0: for non-principal χ there is no pole term at all. Meanwhile the real-pair perturbation at $\delta = \frac{1}{2}$ is *not* zero for χ (largest entry 2.673477). On the ζ side a pole-like object at $z = \mp i/2$ is already present in the form; on the χ side there is nothing of the kind to cancel against a real zero.

4 The function-field analogue

[T1, exact over $\mathbb{Q}$] The same configuration exists in this repository's function-field control, where RH is Weil's theorem. There a real reciprocal pair $\{\rho, \rho^{-1}\}$ gives the rank-two Toeplitz matrix of Theorem 6 of the δ^2–L^3 paper [4],

$$T_R = uw^{\mathsf{T}} + wu^{\mathsf{T}}, \qquad u_i = \rho^i, \quad w_i = \rho^{-i}, \quad 0 \le i \le R.$$

With J the flip $x_i \mapsto x_{R-i}$ one has $Ju = \rho^R w$ and $Jw = \rho^{-R}u$, so the parity directions are the **ρ^R-weighted** vectors $v_\pm = u \pm \rho^R w$, and

$$T_R = \frac{1}{2\rho^R}\left[v_+ v_+^{\mathsf{T}} - v_- v_-^{\mathsf{T}}\right], \qquad T_R v_\pm = \lambda_\pm v_\pm, \qquad \lambda_\pm = (R+1) \pm \rho^{-R}\|u\|^2.$$

Verified exactly in `fractions.Fraction`, no floating point, for $\rho \in \{2, \frac{3}{2}, \frac{5}{4}, \frac{7}{3}, 1\} \times R \in \{1, 2, 4, 8, 16\}$ — 25 cells, 0 failures: the splitting identity, both eigenvector relations, v_+ palindromic and v_- anti-palindromic under $i \mapsto R - i$, $\lambda_+ > 0$, and $\lambda_- \le 0$ with equality iff $\rho = 1$. The negative eigenvalue, exactly:

ρ	$R = 1$	$R = 2$	$R = 4$
2	$-1/2$	$-9/4$	$-261/16$
3/2	$-1/6$	$-25/36$	$-5125/1296$
5/4	$-1/20$	$-81/400$	$-168561/160000$
7/3	$-16/21$	$-1600/441$	$-6088000/194481$
1	0	0	0

($R = 8$ and $R = 16$ are verified in the same run; their exact rationals run to 22 digits in the denominator and are not printed.)

Remark 8 (a correction, carried over). The brief that commissioned the parent extension, and the brief that commissioned this note, both state that the eigenvalues $(R+1) \pm \|u\|\|w\|$ are attained on $u + w$ and $u - w$, with $u + w$ palindromic. That is **false for every** $\rho \ne 1$: it holds in exactly 5 of the 25 cells, which are precisely the degenerate $\rho = 1$ ones, and $T_R(u + w)$ is parallel to $u + w$ only when $\rho^{2R} = 1$. The missing factor is ρ^R. The *polarisation identity* $T_R = \frac{1}{2}[(u+w)(u+w)^{\mathsf{T}} - (u-w)(u-w)^{\mathsf{T}}]$ is nevertheless exact in all 25 cells. Neither correction touches any result of [4], whose proof derives the eigenvalues basis-independently and never names $u \pm w$ as eigenvectors.

What the control shows. The J-even direction carries the positive rank-one piece and the J-odd direction the negative one: the same signed split, with the same attachment of sign to parity, as $+2A_\delta^2$ / $-2S_\delta^2$. So the sign structure is intrinsic to "a real pair in a finite window" and is not an artefact of the archimedean bookkeeping of $L(s,\chi)$. This is a check at a level the leading-constant dictionary of [4] did not reach: that dictionary matched a δ^2 coefficient between the two sides, whereas what matches here is the *sign attached to a parity class*, an exact and discrete datum. It does **not** transfer the consequence of Corollary 2: in the function field RH is a theorem, so there is no configuration to exclude.

5 Consequences for even-sector techniques

[T3] This section is a scope statement about what certain methods can detect. It is not a criticism of any result, and none of the work cited claims otherwise.

Theorems 7, 8 and 11 of the δ^2–L^3 paper [4] are labelled even-sector in their own statements, and that paper says so globally. By Corollary 2, the even sector is precisely the one that cannot respond to a real zero: a positivity statement there is compatible with a real zero at any $\delta \in (0, \frac{1}{2})$. So each of those three is silent on the Siegel configuration *by construction* — not for want of window length, basis size or precision, but because the perturbation it would have to see is positive semidefinite in that sector. The same applies to the four-mode certified witness of Theorem 11, whose witness consists of even quarter-wave modes.

The Connes–van Suijlekom pipeline [2] enters the same way at one specific point: its hypothesis concerns a simple lowest eigenvalue with an *even* eigenfunction, so the quantity it is about is an even-sector ground state. Corollary 2 says that quantity cannot move in response to a planted real zero. That hypothesis has since been certified at one scale — Zhu's Theorem 6.2 puts $\lambda_1^{\mathrm{even}} \le 2.523 \times 10^{-16}$ against $\lambda_2^{\mathrm{even}} \ge 2.085 \times 10^{-12}$ and $\lambda_1^{\mathrm{odd}} \ge 8.206 \times 10^{-15}$ at $L = 0.8$, so the ground state there is simple and even [6] — which settles that step of the programme and leaves the present point untouched: a quantity that is certified, and an even-sector quantity that cannot respond to a real zero, are consistent descriptions of the same number.

Zhu's certificate is over the full space — and that changes nothing, for a reason worth stating. Zhu's certificate is **not** an even-sector case. It is proved for arbitrary complex test functions:

> [quoted] "For every complex $f \in L^2(\mathbb{R})$ with $\operatorname{supp} f \subseteq [-0.8, 0.8]$, $Q(f) \ge 8.9 \times 10^{-18}\|f\|_2^2$. … with no parity or reality restriction on f." [6, Cor. 6.3]

It is assembled from a parity splitting, the even-sector theorem, and a separate odd-sector certificate $Q(f) \ge 8.2065 \times 10^{-15}\|f\|_2^2$ for real odd f on the same window [6, Lem. 6.1, Thm. 1.2, eq. (14)]. **Version 1 of this note left that domain open; it is settled, and settled in the direction version 1 called the more interesting one.**

The consequence version 1 attached to that outcome was wrong, and is withdrawn. It said that a full-space infimum "does constrain" a real zero, because the full space contains the odd sector where the response is $-2S_\delta(f)^2$. That does not follow. By the explicit formula the certified quantity is the sum over *all* zeros, so if a real pair $\{\frac{1}{2} \pm \delta\}$ were present then for odd f

$$Q_\zeta(f) = \sum_{\text{remaining } \rho} |\widetilde{F}(\gamma_\rho)|^2 - 2S_\delta(f)^2 \;\ge\; 8.9 \times 10^{-18}\|f\|_2^2$$

says only that $\sum_{\text{remaining}} \ge 8.9 \times 10^{-18}\|f\|_2^2 + 2S_\delta(f)^2$, which is satisfiable for every δ. The remaining-zero sum is not independently bounded, and $\sum_{\text{remaining}} = Q_\zeta(f) + 2S_\delta(f)^2$ is an identity, so the certificate cannot bound the one term that would have to be bounded. Zhu states the same thing in his own terms: a proof of $\lambda^*(L) > 0$ at a given L is "a finite fragment of RH" [6, §1.1], while only a *negative* value would disprove RH.

So the scope statement is simpler than version 1 made it, and it does not depend on anyone's parity domain:

> **A finite-window positivity certificate carries no information about whether a real zero exists, whatever sector it is proved on.**

Even-sector blindness (Corollary 2) is a sharper and different statement: it is about the *response* of the form to a real zero, it is exact in δ, and it holds at every L and in every basis. The two should not be conflated. Corollary 2 is why an even-sector certificate is silent *by construction*; the paragraph above is why a full-space one is silent as well, for the more elementary reason that positivity on a bounded window is a fragment of the conclusion rather than a constraint on its failure.

Where a window-based statement about real zeros would have to come from. From a test function with a nonzero odd component, and from nothing else: by Remark 5 a real pair contributes $2(A_\delta(f_{\mathrm{e}})^2 - S_\delta(f_{\mathrm{o}})^2)$, so lowering the form requires the odd part to dominate. The odd sector at small L is where that is easiest, the response there being negative and largest relative to the ordinate (Corollary 4). But the route is subject to the obstruction the companion note [3] measured on the Davenport–Heilbronn function: there, the negativity produced by an off-line orbit was partly cancelled by the remaining spectrum.

Two different bounds on the remaining-zero form are involved, and version 1 of this note named the wrong one here. To attribute an *observed* negativity to one planted orbit — the companion note's inference — one needs a *lower* bound on the remaining-zero form. To turn a certified *positivity* into an exclusion of a real zero, one needs an unconditional *upper* bound on it, independent of Q: only then does $\sum_{\text{remaining}} < \lambda^* \|f\|_2^2 + 2S_\delta(f)^2$ contradict the certificate. Neither bound is produced here, and no such upper bound is known to the author for ζ or for any Dirichlet L-function (non-claim 4) — a statement about what is available, not a proof that none exists.

6 Illustration

[T2-N; every row] Smallest scanned L at which a planted real zero at each δ produces a negative odd-sector Rayleigh quotient, at fixed basis dimension $N = 12$, over $L \in \{0.5, 0.8, 1.1, 1.4, 1.7, 2.0, 2.3, 2.6, 3.0\}$ (270 rows, 228 detections):

	$\delta = 0.1$	$\delta = 0.25$	$\delta = 0.375$	$\delta = 0.45$	$\delta = 0.49$
ζ	0.5	0.5	0.5	0.5	0.5
χ_{-3}	0.8	0.8	0.8	0.8	0.8
χ_{-4}	1.1	1.1	1.1	0.8	0.8
χ_5	1.1	1.1	0.8	0.8	0.8
χ_{-7}	1.4	1.1	1.1	1.1	1.1
χ_8	1.4	1.1	1.1	1.1	1.1

So: **odd sector, by $L = 1.4$ with a 12-dimensional basis, for every character and every $\delta \geq 0.1$**; as early as $L = 0.5$ for ζ and $L = 0.8$ for χ_{-3}. And **even sector: never**, at any L, any N, any δ — in the 180 even-sector rows of the Phase-5 scan, 180 have response ≥ 0 and 0 are detections, while all 180 odd-sector rows have response < 0 and 135 are detections. That is Theorem 1 as a prediction, confirmed, not as evidence for itself.

Remark 9 (two caveats that change how the table should be read). First, the ζ row is nearly vacuous: the odd-sector baseline there is $\lambda^*_{\mathrm{odd}}(\zeta, 0.8) = 4.98077 \times 10^{-14}$, so ζ's odd form is already within 10^{-14} of singular and *any* negative rank-one perturbation detects. The informative rows are χ_{-4} and χ_5 at $L = 0.8$, which clear baselines of 0.0988 and 0.0679 respectively (both at

$N = 12$). Second, the ordering of the onsets is the ordering of the odd-sector baselines, and across these five characters the conductor q and the first ordinate γ_1 are not independently varied, so the ordering is not evidence of a conductor effect.

7 Origin of this result

[T3] Recorded because the mechanism turns out to be old, so provenance matters more here than in a note that claims something new.

The trigger. The October 2026 discussion of Siegel zeros, which made $\gamma = 0$ the ordinate worth planting a zero at — the companion work had until then planted only off-line quartets at $\gamma \neq 0$, where the sectors do not separate.

The author's prior inputs. The orbit contribution $4(|P|^2 - |S|^2)$ of the companion note [3, eq. (4)], of which this note is the $\gamma = 0$ degeneration; the window form and its independently derived pole term from the parent assembly [4]; and that paper's function-field control, after Hallouin–Perret [5], which supplied the exact-over-$\mathbb{Q}$ check of §4.

The literature inputs, all three identified *after* version 1 was deposited: Bombieri [1, §9] for the phenomenon itself; Zhu [6, Lem. 6.1] for the $\delta = \frac{1}{2}$ instance, derived independently and in print first, and [6, Cor. 6.3] for the certificate's domain; Connes–van Suijlekom [2] for the even-ground-state hypothesis that §5 discusses.

The AI role. AI assistance was used for the algebra, the numerical verification, the adversarial review, the literature retrieval and the drafting, and both corrections recorded in Appendix B as R7 and R8 were made by reading primary sources rather than summaries of them. Conceptual direction, interpretation and final responsibility are the author's.

8 What this rules out, and what it does not

Ruled out. Reading any even-sector finite-window positivity result — at any L, in any basis, to any precision — as evidence against a real zero near 1. By Theorem 1 the perturbation such a result would have to detect is positive semidefinite in that sector, so the computation is not insensitive to the configuration but *anti*-sensitive to it: planting the zero improves the very quantity being certified.

Not ruled out, and not claimed. Whether any real zero near 1 exists, for ζ or for any $L(s, \chi)$: nothing here bears on it. An odd-sector certificate: none is given here, and none is known for any Dirichlet window — and as §5 shows, one would not suffice anyway, so the odd-sector results are detector-sensitivity measurements and not exclusions. Complex characters: the whole analysis assumes χ real, since it is the reality of χ that makes M_χ real symmetric and the parity split available. And the masking obstruction of the companion note is untouched — this note says where a real zero would be visible, not that it could be isolated from the rest of the spectrum once it was.

A The proof, with the factorisation written out

The content of Theorem 1 is that the autocorrelation factorises. Writing it out once, with every step:

$$\text{zero side} = \sum_{\rho} h(z_\rho), \qquad h(z) = F(z)F(-z), \qquad \rho = \tfrac{1}{2} + iz,$$

$$\begin{aligned}
\text{real pair } \{\tfrac{1}{2} \pm \delta\} \;&\longrightarrow\; z = \mp i\delta && \text{(the functional equation forces the pa}\\
\text{contribution} &= h(i\delta) + h(-i\delta) = 2h(i\delta) && (h \text{ even})\\
&= 2\,F(i\delta)\,F(-i\delta) && \text{(the factorisation — the whole conten}\\
F(i\delta) &= \int f(x)e^{-\delta x}dx = \int f\cosh(\delta x)dx - \int f\sinh(\delta x)dx\\
f \text{ even}: &\quad A_\delta - 0 = A_\delta, \qquad F(-i\delta) = A_\delta + 0 = +A_\delta\\
f \text{ odd}\ : &\quad 0 - S_\delta = -S_\delta, \qquad F(-i\delta) = 0 + S_\delta = +S_\delta
\end{aligned}$$

so the product is $+A_\delta^2$ for even f and $-S_\delta^2$ for odd f, and the contribution is twice it. In matrix form, with $a_k = \int \varphi_k \cosh(\delta u)du/\|\varphi_k\|$ and $s_k = \int \psi_k \sinh(\delta u)du/\|\psi_k\|$ on the even and odd bases, the perturbations are $+2aa^{\mathsf{T}}$ and $-2ss^{\mathsf{T}}$: rank one, of the stated signs.

Two points worth isolating. The sign is *not* a convention: it is the parity of F at a purely imaginary argument, and $F(\pm i\delta)$ are both real for real f of definite parity, with the *same* sign in the even case and *opposite* signs in the odd case. And the statement is exact in δ: nothing is expanded, so there is no $\delta \to 0$ regime in which it fails.

B Retraction log

Reproduced from the parent extension's §0.2, verbatim in substance, plus this session's.

R1. `src/regression_zeta.py` asserted the ζ-side values against the parent's committed CSVs at relative tolerance 10^{-20} and reported 14 failures. **Retracted: the tolerance was wrong, the values are right.** The parent prints eigenvalues to twelve significant digits, so a full-precision recomputation can agree with the stored *decimal string* only to about 10^{-12}; every "failure" sat at $4 \times 10^{-15} \ldots 1.7 \times 10^{-12}$. Tolerance corrected to 5×10^{-12}. No computed value changed. The binding precision on any regression against this repository's CSVs is the CSVs' own.

R2. `src/phase2_gateA_primeflip.py` built $L =$ `mpf('0.8')` *before* setting the working precision, so L carried only 15 digits into a 40-digit run. **Fixed.** The affected quantities were control ratios of order 10^4, so no reported conclusion moved, but the ordering was wrong and is recorded rather than quietly repaired.

R3. `src/planting.py::quartet_block` initially omitted the factor $h_{\text{sign}} = -1$ for the odd basis, where the repository's F returns F/i. **Fixed before use**, caught by the planting self-test (quartet$(\gamma, 0) = 2 \cdot$ pair(γ) failed by 4.04). No result was computed with the bad version.

R4. A commit message listed a fourth retraction — the ζ Connes window — but the row was never added to the table. **Fixed.** The substance: that file's fourth window is $L = \log(13)/2$ stored rounded to 10 digits, and $\lambda_{\min} \approx 10^{-29}$ there is hypersensitive because $2L$ sits exactly at the $n = 13$ prime-power cutoff, so reconstructing L from the stored string moved the answer by 1.8×10^{-8} relative. *The documentation gap is itself the retraction: a commit message is not the record.*

R5. `src/verify_siegel_parity.py` first asserted a stored column at relative tolerance 10^{-25}, which would have reported 12 spurious failures at observed deviations of $4 \times 10^{-11} \ldots 1.3 \times 10^{-10}$.

Fixed before the run that produced the committed transcript: the stored column carries ten significant digits, so the binding tolerance is the file's. *This is R1 recurring — the lesson did not transfer the first time.*

R6 (this session). Two claims in the brief that commissioned this note were repeated from an earlier brief after the parent extension had already refuted them, and are **not** asserted here: that the function-field eigenvalues are attained on $u \pm w$ with $u + w$ palindromic (false for every $\rho \neq 1$; Remark 8), and that Zhu's certificate is one of the even-sector cases that carry no information about the Siegel configuration (not established either way; §5). Both are corrected in place rather than carried forward.

R7 (version 2). Zhu's domain, and a consequence of my own. The paper was obtained and read after version 1 was deposited. Four corrections follow, and two are mine rather than the brief's.

- The domain is the **full admissible space**, with no parity or reality restriction on f [6, Cor. 6.3]. Version 1's "this note does not settle it" is superseded; the brief's even-sector listing was wrong, as R6 said, but for a reason R6 did not have.
- **Withdrawn:** version 1's statement that a full-space infimum "does constrain" a real zero. It does not. The certified quantity is the sum over all zeros, and the real pair's negative odd-sector contribution is absorbed by a remaining-zero sum that nothing here bounds; see §5. The replacement statement is weaker, cleaner, and does not depend on a parity domain at all.
- **Corrected:** "an exclusion needs an unconditional *lower* bound on the remaining-zero form" (§5). For this inference it is an *upper* bound; the lower bound belongs to the companion note's different inference. Both are now named.
- Non-claim 4 said no Zhu-style certificate *exists* for any Dirichlet window. One exists for ζ, in both sectors [6, Thm. 6.2]; for Dirichlet windows the statement stands, now phrased as what is known rather than what exists.

Nothing in §2–§3 changes: the theorem and both corollaries are about the response of the form to a planted real zero, and do not depend on any certificate's domain.

R8 (version 2). Novelty positioning corrected. A post-deposit literature search identified Bombieri [1, §9] as prior art for the sign-by-parity phenomenon itself: Theorem 9 counts the negative eigenvalues of the even and odd sectors by $\Re(\gamma) > 0$ and $\Re(\gamma) \geq 0$ respectively, so a real zero — a conjugate pair with $\Re(\gamma) = 0$ — is counted by the odd sector alone. Version 1 presented the phenomenon without that attribution. It is now attributed, no priority is claimed for it, and what the note adds is stated explicitly after Remark 6. The paper was already cited by this repository's parent work for two other results, and that work had already recorded that Bombieri "separates even and odd eigenfunctions": the information was within reach and was not followed up. *Both defects of version 1 were found by reading a paper already in the bibliography.*

C Reproduction

Code and data: `https://github.com/innerlightr-wq/weil-window-control`, directory `extensions/dirichlet`
Python 3.14.7 (CPython), `mpmath` 1.3.0; no `numpy`, no `float64` on any path to a reported number;
§4 uses `fractions.Fraction` only.

Result	Script	Data
§3 checks (a), (b), (c)	`src/verify_siegel_parity.py`	`data/verify_siegel_parity.csv`, `results/verify_siegel_pari`
§2 the K identity	`src/phase4b_sector_extremal.py`	`data/phase4b_sector_extremal.csv`
§4 function field	`src/ff_parity_split.py`	`results/ff_parity_split.json`
§6 onset, odd sector	`src/phase5_onset.py`	`data/phase5_onset_odd.csv`
§6 even/odd scan	`src/phase5_siegel.py`	`data/phase5_siegel.csv`

`make verify`, in the note directory, runs `note/verify_note_numbers.py`: it rebuilds every quantity printed above from the data files, scans this document for numeric literals, fails if any literal is not derived or is printed to more significant digits than its source carries, regenerates the five machine-generated tables and compares them, and re-checks that the three prerequisite verifications are recorded as passed. `make` builds the PDF. The full extension suite is `./run_suite.sh`.

Licence

As the parent repository: code MIT, this note and the data CC BY 4.0. CC BY 4.0 requires attribution; if you use the note or the data, please cite it by its DOI and the parent paper [4] separately.

Acknowledgments

AI assistance was used for algebra checking, numerical verification, adversarial review, literature organization, and manuscript drafting. All conceptual development, research direction, interpretation, and final responsibility for the content belong to the author.

References

[1] E. Bombieri, *Remarks on Weil's quadratic functional in the theory of prime numbers, I*, Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei (9) Mat. Appl. **11** (2000), no. 3, 183–233; §9 (p. 215), Theorem 9 (p. 217), purely imaginary case (p. 218).

[2] A. Connes and W. D. van Suijlekom, *Quadratic forms, real zeros and echoes of the spectral action*, arXiv:2511.23257.

[3] E. De Jesús, *A Certified Negative Finite-Window Weil Witness for the Davenport–Heilbronn Function, with the Remaining-Zero Form Positive*, Zenodo, 2026. doi:10.5281/zenodo.23608288.

[4] E. De Jesús, *A δ^2–L^3 Sensitivity Law for Finite-Window Weil Positivity, with a Function-Field Control of the Connes–van Suijlekom Pipeline*, Zenodo, 2026. doi:10.5281/zenodo.23250932.

[5] E. Hallouin and M. Perret, *A unified viewpoint for upper bounds for the number of points of curves over finite fields via Euclidean geometry and semi-definite symmetric Toeplitz matrices*, Trans. Amer. Math. Soc. **372** (2019), no. 8, 5409–5451.

[6] X. Zhu, *Weil positivity in compact windows: a finite reduction, certified two-sided bounds, and a Landau–Widom decay law*, arXiv:2608.24827v2, 3 September 2026. The version matters: v2 retracts a support-2.38 claim of an earlier draft.